

First-Principles Study of the Elastic Properties of Germanium Under Directional Strain

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1 Abstract

This report presents a computational process to determine Young's modulus and Poisson's ratio for germanium. The self-consistent field (SCF) relaxed lattice parameter and converged kpoints and energy cutoff were used from a previous bulk modulus computation, and an updated eight-atom conventional unit cell was created in the Quantum ESPRESSO input file. Next, tensile and compression tests were prepared to simulate variable levels of strain along the z-axis, ranging from -5.0 to +5.0 percent. The resulting stress tensor and fractional lattice parameters were extracted from each output file to analyze the relationship between stress and applied strain along the linear axis, and compare the resulting lateral strain due to applied linear strain. Finally, the data from the tensile and compression tests were fitted to linear curves with the slopes of the stress-strain and lateral-linear strain relationships equal to Young's modulus and Poisson's ratio, respectively. The linear fit of the tensile test produced values of 86 GPa for Young's modulus and 0.22 for Poisson's ratio, while the compressive test resulted in values of 81 GPa and 0.30 for the same properties. Comparison of these results to other published computational data and percent error are discussed.

2 Introduction

Young's modulus is a mechanical property of materials that describes of how matter changes shape under tensile or compressive stress [1]. Mathematically, it is represented as

$$Y = \frac{\sigma}{\epsilon} \quad (1)$$

where Y is Young's modulus, σ is the force applied (stress) and ϵ is the strain, or degree of deformity. In this experiment, the strain was varied as a percentage of the original lattice parameter, and stress was extracted from the output files as the pressure along each axis.

Poisson's ratio describes the distortion along each axis due to applied strain, and is evaluated as the negative ratio of the lateral strain to the linear strain [2].

$$-K = \frac{\epsilon_{\perp}}{\epsilon_{\parallel}} \quad (2)$$

In the above equation, K is Poisson's ratio, $\epsilon_{\perp}$ is the strain along the lateral axis, and $\epsilon_{\parallel}$ is the strain along the axis where the stress is applied (linear axis). The lateral strain was determined by evaluating the normalized change in lattice parameter along each unit vector.

For this experiment, the dataset should produce linear relationships between the stress-strain and lateral-linear strain plots, with Young's modulus and Poisson's ratio representing the $\pm$ slopes of each fit.

3 Methods

3.1 Crystal Structure and Lattice Parameters

The SFC-relaxed lattice parameter of germanium from Project 1 was used as the cell parameter for each input file of the tensile and compression tests. The converged k-points and energy cutoff of 12 and 110 Ry, respectively, from 1 were also used.

A conventional unit cell of germanium was constructed with eight atoms as opposed to the two-atom primitive unit cell that was used in Project 1. The conventional cell allows the unit cell vectors to align orthogonally, which simplifies the tensile test [3]. Figure 3.1 shows the starting coordinates for the initial relaxation, with no strain applied to the crystal structure.

```
CELL_PARAMETERS alat
1.0 0.0 0.0
0.0 1.0 0.0
0.0 0.0 1.0

ATOMIC_POSITIONS alat
Ge 0.00 0.00 0.00
Ge 0.25 0.25 0.25
Ge 0.00 0.50 0.50
Ge 0.25 0.75 0.75
Ge 0.50 0.00 0.50
Ge 0.75 0.25 0.75
Ge 0.50 0.50 0.00
Ge 0.75 0.75 0.25
```

Figure 3.1 – Initial cell parameters and atomic positions, 0 strain.

3.2 Edit Input File

The Quantum ESPRESSO input file was updated to define the system for the tensile and compression tests. Key parameters were defined as:

- `celldm` = 10.879306357, the relaxed unit cell parameter in units of Bohr
- `nat` = 8, the number of atoms in the conventional unit cell
- `ecutwfc` = 110, the maximum energy plane waves are allowed to have in units of Rydberg
- `nbnd` = 73, the number of electron bands to calculate
- `cell_dofree` = 'xy', allows only the x and y unit vectors to relax [4]
- `CELL_PARAMETERS` = [1.0 0.0 0.0], [0.0 1.0 0.0], [0.0 0.0 1.0], this was changed iteratively to simulate different degrees of strain
- `ATOMIC_POSITIONS` = [0.00 0.00 0.00], [0.25 0.25 0.25], [0.00 0.50 0.50], [0.25 0.75 0.75], [0.50 0.00 0.50], [0.75 0.25 0.75], [0.50 0.50 0.00], [0.75 0.75 0.25], the fractional coordinates of the atoms in the unit cell
- `K_POINTS` = 12 12 12, the number of points in a periodic mesh where the computer measures the electrons in a crystal structure

The number of bands (`nbnd`) was adjusted to 73, using the typical method of dividing the total number of valence electrons in the conventional cell by 2 and adding 30% for margin. While germanium has four true valence electrons, 2 in the *4s* state and 2 in the *4p* state, the electrons from the *3d* state are included because they are high enough in energy to affect bonding and can overlap with valence states of neighboring atoms [5]. The use of 14 valence electrons was verified by the `Z valence = 14` parameter in the pseudopotential file.

3.3 Perform Tensile and Compression Tests

Input files were created to simulate varying levels of strain applied to the z-axis. The z-coordinate of `CELL_PARAMETERS` was updated in each input file to a specific strain: 1.01, 1.02, 1.03, 1.04, and 1.05 for tension and 0.99, 0.98, 0.97, 0.96, and 0.95 for compression.

As the level of strain increased, simulation time-outs became an issue. The maximum allowable runtime for student accounts in Wendian is 4 hours, which was not enough to run all jobs to completion. As a result, the number of tasks per node was increased from 4 to 8 in the `sbatch.sh` file.

Increasing the number of tasks per node improved the successful completion of jobs, but some still timed out. To resolve this, the `restart_mode` parameter in the input file was changed to 'restart' and the issue was resolved. The 'restart' setting starts jobs from the end of the previously interrupted run, significantly reducing the number of iterations needed to reach convergence [4].

3.4 Analyze Stress-Strain and Lateral-Linear Strain Relationships

From each output file, the pressure on each axis was recorded in units of kbar, and the final cell parameters, which represent a fraction of the lattice parameter, were also recorded. Figure 3.2 shows the stress tensor and final cell parameters for the +1.04 strain tensile test.

(kbar)			P=	-11.58	CELL_PARAMETERS (alat= 10.87930635)		
-0.14	0.00	0.00		0.00	0.990936314	0.000000000	0.000000000
0.00	-0.14	0.00		0.00	0.000000000	0.990936314	0.000000000
0.00	0.00	-34.45			0.000000000	0.000000000	1.040000000
(a)					(b)		

Figure 3.2 – Output file stress tensor (a) and final cell parameters (b) for the +1.04 tensile test.

The data was organized in a table by percent strain applied to the z-axis. Stress data was converted from kbar to GPa by dividing the value by 10. The non-zero cell parameters were recorded and used to scale the lattice parameter along each axis. Finally, the lateral and linear strain for each run was calculated as shown in equations 4 and 5. Figure 3.3 is the resulting data chart from this exercise.

	0.95	0.96	0.97	0.98	0.99	1.00	1.01	1.02	1.03	1.04	1.05
strain	-0.05	-0.04	-0.03	-0.02	-0.01	0.00	0.01	0.02	0.03	0.04	0.05
z P (kbar)	-40.45	-32.85	-25.05	-17.16	-8.69	-0.07	8.74	17.23	25.96	34.45	43.04
z P (GPa)	-4.045	-3.285	-2.505	-1.716	-0.869	-0.007	0.874	1.723	2.596	3.445	4.304
z celldm factor	0.95	0.96	0.97	0.98	0.99	1.00	1.01	1.02	1.03	1.04	1.05
z celldm (Bohr)	10.3353	10.4441	10.5529	10.6617	10.7705	10.8793	10.9881	11.0969	11.2057	11.3145	11.4233
Δz (Bohr)	-0.5440	-0.4352	-0.3264	-0.2176	-0.1088	0.0000	0.1088	0.2176	0.3264	0.4352	0.5440
$\epsilon_{ }$	-0.05	-0.04	-0.03	-0.02	-0.01	0.00	0.01	0.02	0.03	0.04	0.05
xy P (kbar)	0.34	0.26	0.19	0.46	0.30	0.07	-0.39	-0.09	-0.12	-0.14	-0.15
xy P (GPa)	0.034	0.026	0.019	0.046	0.030	0.007	-0.039	-0.009	-0.012	-0.014	-0.015
xy celldm factor	1.015	1.012	1.008	1.005	1.003	1.000	0.998	0.995	0.993	0.991	0.989
xy celldm (Bohr)	11.0398	11.0049	10.9710	10.9360	10.9067	10.8793	10.8555	10.8275	10.8033	10.7807	10.7591
Δxy (Bohr)	0.1605	0.1256	0.0917	0.0567	0.0274	0.0000	-0.0238	-0.0519	-0.0760	-0.0986	-0.1202
$\epsilon_{\perp}$	0.0148	0.0115	0.0084	0.0052	0.0025	0.0000	-0.0022	-0.0048	-0.0070	-0.0091	-0.0110

Figure 3.3 – Stress, lattice parameter, and directional strain data from tensile and compression tests.

$$\epsilon_{\perp} = \frac{\Delta xy}{xy} \quad (4)$$

$$\epsilon_{||} = \frac{\Delta z}{z} \quad (5)$$

3.5 Linear Fitting to Extract Young's Modulus and Poisson's Ratio

3.5.1 Young's Modulus

The stress-strain data from Figure 3.3 was plotted to fit a linear curve. Results from the tensile and compression tests were plotted separately to analyze the agreement of Young's modulus between them, then the data from both tests was combined to plot a single curve. Figure 3.4 depicts the three plots and analysis of each linear fit.

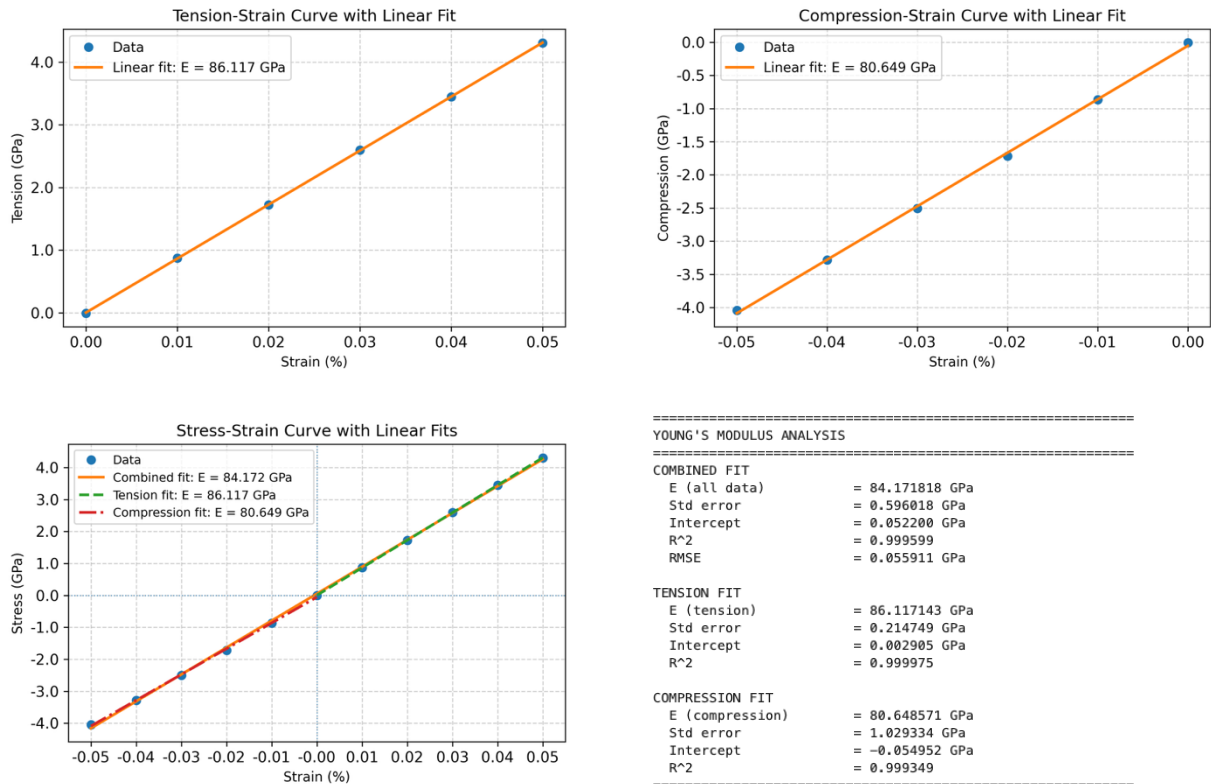


Figure 3.4 – Linear fits of tensile and compression tests, and fit comparison and analysis. The slope of each line is Young's modulus.

Young's modulus was calculated as 86.12 GPa from the tensile test, 80.65 GPa from the compression test, and 84.17 GPa from the combined data. The analysis suggests an excellent linear fit from the data.

3.5.2 Poisson's Ratio

A similar procedure was followed to compare the lateral and linear strain and determine Poisson's ratio. Figure 3.5 depicts the three plots and analysis of the linear fit.

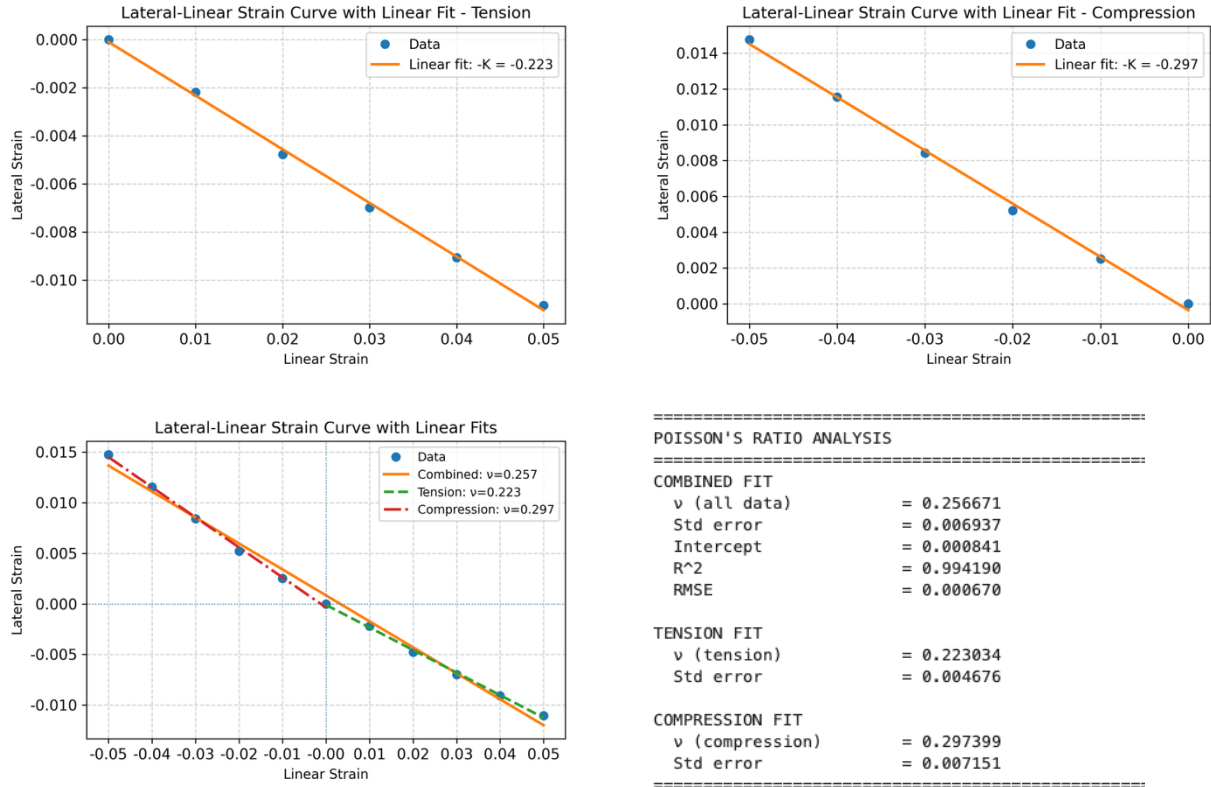


Figure 3.5 – Linear fits of tensile and compression tests, and fit comparison and analysis. The slope of each line is negative Poisson's ratio.

Poisson's ratio was calculated as 0.223 from the tensile test, 0.297 from the compression test, and 0.257 from the combined data. The analysis shows a slightly poorer linear fit than the Young's modulus fit, but still has sound accuracy.

4 Results and Discussion

4.1 Comparison to Literature

While some sources report Young's modulus and Poisson's ratio explicitly, many of them reported stiffness tensor components (c_{11} , c_{12}). The stiffness tensor is related to Hooke's law, and describes the linear relationship between stress and strain. Hooke's law is classically represented by the equation

$$F = -kx \quad (6)$$

where F is the force, x is the displacement from equilibrium, and k is the elastic constant. In this experiment, k is represented by the stiffness tensor that describes the stress-strain along each unit vector. From this relationship, Young's modulus in the [100] orientation can be calculated from the tensor components as

$$Y_{[100]} = \frac{(c_{11}-c_{12})(c_{11} + 2c_{12})}{(c_{11} + c_{12})} \quad [6][7] \quad (7)$$

and Poisson's ratio as

$$\nu_{[100]} = \frac{c_{12}}{c_{11} + c_{12}} \quad [8][9]. \quad (8)$$

There is a wide range of Young's modulus computations among published results with 103 GPa [9] widely cited in material properties databases. Comparing this figure to the experimental results in this paper, the percent error is 16.5% for the tensile test and 21.4% for the compression test. It is important to note, however that the 103 GPa value was produced by means of physical experimentation with germanium and

not computational methods. Results are available from other DFT experiments, but there is significant dispersion of results depending on the exchange correlation used, ranging from 68 GPa to 132 GPa [10] [11] [12] [13] and corresponding relative errors of 0.97% to 36.13%.

While the range of reported Poisson's ratio values is narrower, minor deviations elicit significant error due to the property's order of magnitude. Reviewing literature values, Poisson's ratio is reported between 0.19 and 0.37 [10] [11] [12] [13] and corresponding relative errors of 1.28% to 35.09%. Figure 3.6 shows the spread of literature data and corresponding error.

The broad range of reported data means that there are likely many reasons the values don't correlate. Variations in the simulation setup and DFT parameters such as kpoints, energy cutoff, strain range, pseudopotentials, unit cell structure, and the inclusion of Van-der Waals forces can lead to significant error in reported elastic properties [14]. Further experiments could be done to track these variables and determine their direct influence on directional stress-strain simulations. Additional error may have also been introduced when calculating Young's modulus and Poisson's ratio from published stiffness tensor components.

	Young's Modulus (Y)	Poisson's Ratio (v)	Relative Error - Y	Relative Error - v
Experimental	84	0.26		
QuantumATK - PBE-sol	68	0.37	23.78%	30.63%
QuantumATK - pps-PBE	112	0.26	24.85%	1.28%
QuantumATK - PBE	85	0.24	0.97%	6.95%
QuantumATK - HSE	105	0.24	19.84%	6.95%
Materials Project	83	0.19	1.41%	35.09%
Barhoumi - LDA	119	0.23	29.21%	14.08%
Barhoumi - PBE	102	0.21	17.13%	22.22%
Barhoumi - HSE	132	0.21	36.13%	24.60%
Barhoumi - RPA, PBE0	130	0.21	35.43%	22.22%
Wortman	103	0.26	18.28%	1.28%
Abdurrazq HSE	104	0.25	19.07%	2.67%
Average Lit	104	0.24	20.55%	15.27%

Figure 3.6 – Spread of experimental data results and corresponding percent error.

5 Conclusion

In this work, the structural and mechanical properties of germanium were investigated using DFT computations. Tensile and compression tests were executed to simulate positive or negative strain along the z-axis of the crystal by varying degrees and letting the x and y vectors relax. Using this data, Young's modulus was calculated by fitting stress-strain data to a linear curve. Similarly, Poisson's ratio was calculated by fitting lateral-linear strain data to a linear curve. Both parameters were identified as the slope of the line created by fitted data.

The experimental values of Young's modulus and Poisson's ratio, 84 GPa and 0.26 respectively were consistent with some literature, but the spread of published values was too large to glean any significant comparison of the data. These discrepancies may be due to inconsistencies in the DFT parameters such as kpoints, energy cutoff, strain range, pseudopotentials, unit cell structure, and the inclusion of Van-der Waals forces.

6 References

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